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MILITARY SPECIFICATION

RECOVERY ASSIST, SECURING AND TRAVERSING SYSTEM FOR LAMPS MK III HELICOPTER

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, development and test requirements for the helicopter Recovery Assist, Securing and Traversing (RAST) System.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-W-410	Wire Rope and Strand
TT-E-490	Enamel, Silicone Alkyd Copolymer, Semigloss
TT-P-645	Primer, Paint, Zinc-chromate, Alkyd Type

MILITARY

MIL-S-901	Requirement For High Impact Shock Tests For Shipboard Machinery Equipment And Systems
MIL-C-915	General Specification For Electrical Cable And Cord For Shipboard Use
MIL-E-917	Electric Power Equipment, Basic Requirements (Naval Shipboard Use).

Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter).

FSC 1710

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MIL-C-5541	Chemical Conversion Coating On Aluminum and Aluminum Alloys
MIL-S-8660	Silicone Compound
MIL-Q-9858	Quality Program Requirements
MIL-P-15024	Plates, Tags And Bands For Identification of Equipment
DOD-E-15090	Enamel, Equipment, Light-gray (Formula No. 111)
DOD-P-15328	Primer (Wash), Pretreatment, Blue (Formula No. 117 for Metals)
MIL-E-15936	Enamel, Exterior, Gray, No. 27 (Vinyl-alkyd) (Formula No. 122-27)
MIL-C-16173	Corrosion Protective Compound, Solvent Cutback, Cold-application
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship and Shore, General Specification
MIL-M-17060	Motors, 60 Cycle, Alternating Current, Integral Horsepower (Shipboard Use)
DOD-P-17545	Primer Coating, Alkyd Red Lead Type, Formula No. 116 And Formula No. 116D
MIL-C-18255	Caulking Compound, Synthetic Rubber Base, Wooden Deck Seam Application
MIL-N-18307	Nomenclature And Identification For Electronic, Aeronautical, And Aeronautical Support Equipment Including Ground Support Equipment
MIL-T-22361	Thread Compound, Anti-sieze, Zinc Dust petrolatum
MIL-C-24176	Cement, Epoxy, Metal Repair and Hull Smoothing
MIL-F-24402	Filter, Filter Elements (25 Micrometer Absolute), and Filter Differential Pressure Indicators
MIL-P-24441	Paint, Epoxy-polyamide, General Specification For
DOD-P-24555	Paint, Aluminum, Heat Resisting (650 Deg. C), Low Emissivity (0.40 or Less)
MIL-C-45662	Calibration System Requirement
MIL-S-81733	Sealing and Coating Compound, Corrosion Inhibitive

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MIL-V-81940 Hydraulic Sampling and Bleed Valves, Type II Systems

MIL-H-83282 Hydraulic Fluid, Fire Resistant Synthetic Hydrocarbon Base, Aircraft

SD-567-1 Detail Specification for Model SH-60B Helicopter, FY-81 Procurement

STANDARDS

MILITARY

MIL-STD-108 Definition Of and Basic Requirement For Enclosures For Electric and Electronic Equipment

MIL-STD--109 Quality Assurance Terms and Definitions

MIL-STD-167-1 Mechanical Vibrations of Shipboard Equipment (Type 1- Environmental and Type 2- Internally Excited)

MIL-STD-280 Definitions Of Item Levels, Item Exchangeability, Models, and Related Terms

MIL-STD-419 Cleaning and Protecting Piping, Tubing, and Fittings for Hydraulic Power Transmission Equipment

MIL-STD-454 Standard General Requirements For Electronic Equipment

MIL-STD-461 Electromagnetic Interference Characteristics Requirements for Equipment

MIL-STD-470 Maintainability Program Requirements

MIL-STD-471 Maintainability Demonstration

MIL-STD-721 Definitions of Effectiveness Terms For Reliability, Maintainability, Human Factors, And Safety

MIL-STD-777 Schedule Of Piping, Valves, Fittings, And Associated Piping Components For Naval Surface Ships

MIL-STD-785 Reliability Program for Systems and Equipment Development and Production

MIL-STD-794 Parts And Equipment, Procedures For Packaging And Packing Of

MIL-STD-810 Environmental Test Methods

MIL-STD-831 Test Reports, Preparation Of

MIL-STD-965 Parts Control Program

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MIL-STD-1310	Shipboard Bonding, Grounding, And Other Techniques For Electromagnetic Compatibility And Safety
MIL-STD-1353	Electrical Connectors And Associated Hardware, Selection And Use Of
MIL-STD-1378	Requirements For Employing Standard Electronic Modules
DOD-STD-1399 Section 300	Interface Standard for Shipboard Systems, Section 300 - Electric Power, Alternating Current.
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities
MIL-STD-1683	Selection Standard for Shipboard Use Of Electric Connectors And Jacketed Cable
MIL-STD-2074	Failure Classification For Reliability Testing

DRAWINGS

NAEC 700,000	Recovery Assist, Secure and Traverse System

PUBLICATIONS

NAVSEA 0902-001-5000	General Specifications for Ships of the United States Navy, dated 1 January 1980

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SSPC-SP-10	Surface Preparation Specification No. 10

(Application for copies should be addressed to the Steel Structures Painting Council, 1400 Fifth Ave., Pittsburgh, Pa.)

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3. REQUIREMENTS

3.1 System definition

3.1.1 System modes of operation. The system, when installed, shall provide for the following modes of operation with the LAMPS MK III helicopter.

3.1.1.1 Recovery assist. The system shall provide for the safe, efficient attachment of a tethering cable to a hovering helicopter and shall provide the means for application of cable tension. The cable tension shall be maintained within specified limits during the prelanding phase when the ship and helicopter motions vary erratically due to the wind and sea. The cable tension shall be maintained during the landing phase to minimize the dispersion from the designated landing area.

3.1.1.2 Secure. The system shall provide for rapid and positive securing of the helicopter upon landing within the designated area. Security shall be effected and maintained by mechanically engaging a probe extended from underneath the helicopter fuselage. Yaw control of the helicopter shall be provided by a ship-mounted grid which can be engaged by an auxiliary aircraft probe. The system shall also provide for security of the helicopter to and from the hangar parking position.

3.1.1.3 Traverse. The system shall provide for the controlled movement of the helicopter between the designated landing and takeoff position and the hangar parking position. Security of the helicopter shall be maintained throughout the traversing operation.

3.1.1.4 Launch. The system shall provide for the security of the helicopter at the designated take-off position during the pre-flight operations and shall provide for release of the helicopter main probe from a remote control station.

3.2 System descriptions

3.2.1 General system requirements. The RAST system shall be an integration of defined system elements (see 3.2.2) and candidate ship hull structure to provide the specified system requirements.

3.2.2 Detail item requirements

3.2.2.1 Recovery Assist Winch and Hydraulic Power Unit. The Recovery Assist Winch and Hydraulic Power Unit shall be composed of an electrically-powered hydrostatic transmission driving a winch drum. The unit shall also provide hydraulic power to the Traverse Winches and the Tailguide Winch. The unit shall employ MIL-H-83282 hydraulic fluid and contain a hydraulic reservoir of sufficient capacity to permit system filling to a minimum of 125% of system operating requirements for hydraulic fluid. The unit shall contain a control test panel which will permit local control, operation and observation of performance of the unit for maintenance purposes. The prime mover of the unit shall be an electric motor, conforming to MIL-M-17060. The Recovery Assist Winch and Hydraulic Power Unit shall conform to the installation and interface requirements of Installation Control Drawing,

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NAEC 700,000, and, when combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1. The winch drum of the unit shall be compatible with the cable of the Recovery Assist Cable Assembly (see 3.2.2.3); shall possess an effective drum capacity of 95 feet of that Recovery Assist Cable; shall incorporate a friction brake to prevent drum rotation during non-operation of the recovery assist winch; and shall include provisions to positively guide the Recovery Assist Cable onto and off of the drum.

3.2.2.2 Traverse Winch. The Traverse Winch shall be composed of a winch drum powered by a hydraulic motor which receives hydraulic power from the Recovery Assist Winch and Hydraulic Power Unit. The Traverse Winch shall conform to the installation and interface requirements of Installation Control Drawing, NAEC 700,000 and, when combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1. The winch drum shall possess sufficient wire rope capacity to obtain an effective system traverse length of 90 feet; the wire rope shall conform to RR-W-410. The winch shall be capable of exerting a pull force on the traverse cable of 5000 lbs minimum at cable speeds of 3 ± .5 in/sec. and 12 ± 1.2 in/sec when driven by hydraulic power from Recovery Assist Winch and Hydraulic Power Unit. In addition, the winch shall be capable of exerting a pull force of 8400 lbs without stalling. The Traverse Winch shall contain a friction brake that, when activated shall hold a cable tension of 10,000 lbs minimum without slippage.

3.2.2.3 Recovery Assist Cable Assembly. The Recovery Assist Cable Assembly shall consist of 5/16 inch diameter wire rope conforming to RR-W-410 with one end terminating in a swaged fitting compatible with the Cable End Fitting. The Cable End Fitting shall employ a shear pin which shall fail at a cable tension of 6000 ± 500 lbs. The Recovery Assist Cable End Fitting shall be dimensionally and functionally compatible with the RAST messenger cable end fitting, winch locking mechanism and main probe being provided in SH-60B helicopters conforming to the requirements of SD-567-1.

3.2.2.4 Rope Accumulator. The Rope Accumulator shall be composed of a single acting hydraulic cylinder around which the Recovery Assist Cable (see 3.2.2.3) is reeved. When compressed by tension in the Recovery Assist Cable, the hydraulic cylinder shall discharge the hydraulic oil into nitrogen-oil accumulators. The working volume and nitrogen precharge pressure of each accumulator shall be chosen to dampen cable tension excursions during system operations as defined in 3.4.1.1. The Rope Accumulator shall conform to the installation and interface requirements of Installation Control Drawing, NAEC 700,000, and when combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1. The unit shall employ MIL-H-83282 hydraulic fluid.

3.2.2.5 Moveable Sheave Assembly. The Moveable Sheave Assembly shall conform to the installation and interface requirements of Installation Control Drawing, NAEC 700,000, and, when combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1.

3.2.2.6 Electrical Cable Reel. The Electric Cable Reel shall provide for the paying-out and stowage of the electric cable carrying electrical power and

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control signals to, and condition signals from, the Rapid Securing Device. The reel shall have sufficient capacity to accommodate a Rapid Securing Device traverse distance of 90 feet. Electric cable employed shall conform to MIL-C-915. The Electric Cable Reel shall conform to the installation and interface requirements of Installation Control Drawing, NAEC 700,000, and, when combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1. The cable reel shall generate a cable retraction force in accordance with the following minimums and maximum: 40 lbs minimum; 80 lbs minimum when 90 ft cable is extended; and a maximum of 150 lbs under any condition of extension up to 90 ft. Electrical connectors shall be selected in accordance with MIL-STD-1353. Electrical cable supplied with the reel shall be 150 feet in length.

3.2.2.7 Control Console. The Control Console shall contain all of the system controls, indicators, and switches to be employed in the normal operation of the RAST system. The console shall conform to the installation and interface requirements of Installation Control Drawing, NAEC 700,000, and, when combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1. In addition, the console shall provide for effective operator voice communications with the helicopter and other ship stations; the following specific provisions shall be incorporated:

a. A three-position microphone keying switch shall be located on the Recovery Assist Tension Lever. The switch shall be spring-loaded to the center-off position with switch action away from the operator providing keying of the radio channel, and switch action toward the operator providing keying of the interior ship channel.

b. Provisions in the console shall be made for incorporation of Government furnished communications components consisting of one modified AM-6949 enclosure module, one 12 volt dc power supply (5" x 4" x 2" maximum size), one 24 volt dc power supply (5" x 4" x 2" maximum size), one line filter (5" x 4" x 2 " maximum size), one transformer (3" x 4" x 2" maximum size), two headsets with microphones, and associated mating connectors and cables.

3.2.2.8 Tail Probe Grid. Tail Probe Grid components shall conform to the installation and interface requirements of Installation Control Drawing, NAEC 700,000 and to the interface requirements of the SH-60B tail probe conforming to SD-567-1.

3.2.2.9 Tailguide Winch . The Tailguide Winch shall be composed of two winch drums, port and starboard, driven by separate hydraulic motors. Hydraulic power shall be derived from the Recovery Assist Winch and Hydraulic Power Unit. The Tailguide Winch shall conform to the installation and interface requirements of Installation Control Drawing, NAEC 700,000, and, when combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1. The Tailguide Winch drums shall have a usable capacity of 40 feet of wire rope conforming to RR-W-410. Provisions shall be incorporated to limit the tension applied to the tailguide cables, from either winch drum, such that the tension does not exceed 4000 lbs at the drums in any mode of Tailguide Winch operation.

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3.2.2.10 Electronic Components Assembly. Except for sensors and certain switching relays, the Electronic Components Assembly shall provide the centralized assembly of electronic components and circuit elements required to effectively control applicable units of the RAST system in response to command inputs from the Control Console. The Electronic Components Assembly shall conform to the installation and interface requirements of Installation Control Drawing, NAEC 700,000, and, when combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1. The assembly shall employ standard electronic modules (see 3.5.1.1).

3.2.2.11 Rapid Securing Device. The Rapid Securing Device shall be dimensionally and functionally compatible with the interface requirements of the SH-60B helicopter and the RAST main probe installed therein as defined in SD-567-1. In addition the Rapid Securing Device shall conform to the installation and interface requirements of Installation Control Drawing NAEC 700,000, and, when combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1. The Rapid Securing Device shall utilize electrical power in accordance with 3.3.2.3; electrical power and signals to and from the Rapid Securing Device shall be via the electrical cable of the Electric Cable Reel. The Rapid Securing Device shall employ arresting beam assemblies which lock around the aircraft probe when the probe lands anywhere within the rectangular capture area. These arresting beams shall restrict the movement of the probe in the fore-and-aft direction and shall be engaged by a brake system to hold the locked beams in the athwartships direction. The beam brake system shall be releasable by electrical command, under any lateral load up to 3000 lb imposed by the probe, to allow for transverse movement of the locked beams while straightening the aircraft. Following engagement, the beam brakes shall remain engaged under any conditions of electrical power loss or hydraulic failure. The Rapid Securing Device shall contain a self-contained, electrically powered, hydraulic system employing MIL-H-83282 hydraulic fluid for the activation of arresting beam movement, the beam locks, and the beam brake system. The hydraulic system shall contain an accumulator with sufficient capacity to provide 4 complete open and close cycles of operation from a fully charged condition should electric power fail, and shall have a manually operated pump to permit charging the hydraulic system without electric power. The Rapid Securing Device shall contain visual signalling indicator(s) to show that the arresting beams are locked together. The Rapid Securing Device shall be capable of manual control of arresting beam closing and opening from a safe distance using a lanyard.

3.2.2.12 Track system components. Track system components shall conform to the installation and interface requirements of Installation Control Drawing, NAEC 700,000 and the interface requirements of criteria between the RAST system and the SH-60B helicopter as defined in SD-567-1; and, when installed and combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1.

3.2.2.13 Motor starters. Motor starters for the Rapid Securing Device and for the Recovery Assist Winch and Hydraulic Power Unit shall conform to the installation and interface requirements of Installation Control Drawing, NAEC

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700,000, and, when installed and combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1.

3.2.2.14 Traverse Winch Valve Assembly. The Traverse Winch Valve Assembly shall conform to the installation and interface requirements of Installation Control Drawing, NAEC 700,000, and, when installed and combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1.

3.2.2.15 Tachometer Sheave. The Tachometer Sheave shall conform to the installation and interface requirements of Installation Control Drawing, NAEC 700,000, and, when installed and combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1.

3.3 Aircraft and ship compatibility. Compatibility requirements are derived from the basic requirements that the system, when installed on any of the various classes of LAMPS MK III ships, will operate satisfactorily with the LAMPS MK III aircraft.

3.3.1 Aircraft compatibility

3.3.1.1 Aircraft interface criteria. All deck equipment of the system shall be dimensionally compatible for operation with the LAMPS MK III helicopter under both static and dynamic conditions associated with landing and traversing of the aircraft. Units of the RAST system shall conform to the interface requirements of SD-567-1. The Rapid Securing Device shall satisfactorily interface with all normal aircraft configurations with the probe locked in any position within the device.

3.3.1.2 Tire impact. The RSD shall be capable of withstanding, without permanent deformation, the ultimate load of a LAMPS MK III helicopter main wheel landing impact (as defined in SD-567-1) while supported on a track which does not suffer permanent deformation.

3.3.1.3 Tire hazards. The corners and physical projections of the RSD, track, and other deck equipment shall be designed to minimize potential damage to helicopter tires.

3.3.2 Ship compatibility

3.3.2.1 System installation. All components to be supplied under this specification shall meet the size and weight criteria and other installation requirements of the Installation Control Drawing, NAEC 700,000. Individual track, tail probe grid and associated components shall be procured on the basis of their requirement for a particular installation.

3.3.2.2 Hangar door seal interface. The design of the track system shall allow for installation of seals within the track trough at the hangar face to minimize spilled fuel and water entering the hangar or below deck spaces and to maximize light tight integrity.

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3.3.2.3 Electrical power. The system shall utilize 440 volt, 3 phase, 60 Hz, Type I power per DOD STD-1399, section 300, except the RSD which may utilize 400 Hz, Type I power.

3.4 Characteristics

3.4.1 Performance Characteristics. RAST system equipment, when installed in accordance with the installation and interface requirements of Installation Control Drawing, NAEC 700,000 and combined into a system configuration, shall conform to the system performance requirements as defined for the following modes of operation which shall be selectable from the Control Console.

3.4.1.1 Recovery Assist. System operation in the Recovery Assist mode (see 3.1.1.1), using controls located on the Control Console, shall provide the following functional capability within the stated limits:

a. A capability to reel-out the Recovery Assist Cable shall be provided in which the force required at the on-deck end of the cable to pay out cable shall not exceed 50 lbs. Pay out speed need not exceed one foot per second.

b. A capability to reel-in untensioned Recovery Assist Cable shall be provided in which the reel-in speed shall not exceed two feet per second; reel-in shall automatically stop when, or if, tension in the Recovery Assist Cable exceeds 400 lbs.

c. A capability to select and control the tension in the Recovery Assist Cable between a minimum tension of 850 ± 50 lbs.and a maximum tension of 4000 lbs. shall be provided. The control system shall provide control of the Recovery Assist Cable within the following limits.

(1) Under static conditions (a cable speed of zero) the value of tension displayed on the indicator of the Control Console shall be within 100 lbs. of the actual cable tension over the range from 850 ± 50 lbs to 4000 lbs.

(2) Under dynamic conditions, in which the cable is cycled in a sinusoidal manner between zero and specified peak cable speed at 9.5 ± 5% cycles per minute constant frequency cycling, the tension deviations from the mean value shall not exceed the following deviations limits for operations at the indicated nominal mean tension values:

Nominal Mean Cable Tension (lbs)	Peak Cable Speed (ft/sec)	Tension Deviation Limit (± lbs)
850	18	150
1500	18	225
2000	18	300
3000	13.3	400
3500	11.4	400
4000	10	400

(3) Under dynamic conditions, in which the simultaneous operations of selection of minimum tension at the Control Console and extension of the

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Recovery Assist Cable at a speed of 18 ft/sec (i.e., the application of a snatch load), the peak cable tension shall not exceed 1200 lbs.

(4) Under dynamic conditions, in which the Recovery Assist Cable is winched-in with a tension of 4000 lbs at speeds up to 12 ft/sec, the peak tension encountered shall not exceed 5500 lbs when cable movement is stopped (simulating the action of the helicopter landing gear oleo) by compression of three Neoprene slabs between a block clamped to the cable and a fixed stop plate. The Neoprene slabs shall be prepared from material of 60-65 Durometer rating, 2 inches thick and 4 inches square with a 5/16 inch diameter hole centered on the 4" x 4" face for threading the cable.

(5) Under dynamic conditions, in which the Recovery Assist Cable tension is commanded to change from a minimum tension of 850 ± 50 lbs to a tension of 3500 lbs in 0.65 seconds by advancement of the Cable Tension Lever on the Control Console, the cable tension overshoot above 3500 lbs shall not exceed 700 lbs and the tension oscillations shall decay to 3500 ± 50 lbs within 0.85 seconds from the time tension initially reaches 3500 lbs.

3.4.1.2. Secure. System operation in the Secure mode (see 3.1.1.2) shall provide the following capabilities within the stated limits.

a. The system shall be capable of securing the LAMPS MK III helicopter through the Main Probe engaged in the Rapid Securing Device and the Tail Probe engaged in either the Tail Probe Grid or in the Tail Probe track slot, under conditions of the helicopter rotor turning or stopped, and the ship experiencing motions and winds associated with upper sea state five, that generate the following probe load conditions:

(1) Main Probe

Vertical Load - 24,000 lbs.
Lateral Load - 21,000 lbs.
Longitudinal Load - 10,000 lbs.

(2) Tail Probe

Lateral Load - 5,000 lbs.

Main Probe loads are expressed in relation to the axis of the Rapid Securing Device; Tail Probe loads are expressed relative to grid or track slots. All loads are Design Limit or Maximum Working Loads. Equipment design shall include a safety factor of 1.50.

b. Using controls located on the Control Console, the system shall provide for the closure and locking of the Rapid Securing Device Arresting Beams around the aircraft Main Probe within two seconds of control actuation. This locking action shall be of a positive type that requires specific actuation to unlock.

c. The system shall provide for the sensing, transmission and display on the Control Console of the following physical conditions within the selected Rapid Securing Device.

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Arresting beams open
Arresting beams closed and locked together
Beam brakes on
Hydraulic pressure low

d. Security of the helicopter through operation of the Rapid Securing Device or the Traverse Winch and cables shall be positive up to the required load limits when the helicopter probes are engaged. Helicopter security about the yaw axis shall be achievable through attached Tailguide Winch cables which shall be capable of providing restraining forces up to 3400 lbs when the helicopter Tail Probe is not engaged; winch restraining tension shall not exceed 4000 lbs.

e. The RAST system, in restricting the swing of the helicopter tail through the attached Tailguide Cables, shall limit the Tailguide Cable tension at the helicopter attachment point to not more than 4000 lb at cable payout speeds up to 16 in/sec.

f. System design for helicopter security, with the exception of the Traverse Winch brakes, shall not be predicated on friction forces which could be lost by the introduction of lubrication fluids.

3.4.1.3 Traverse. System operation in the Traverse mode (see 3.1.1.3) shall provide the following capabilities within the stated limits.

a. The system shall provide all mechanical forces required to straighten and reposition a helicopter that has been recovered and secured with the Main Probe engaged in the Rapid Securing Device and the Tail Probe engaged in the Tail Probe Grid, and to traverse the helicopter from the landing area to the hangar parking position. Security of the helicopter in accordance with 3.4.1.2 shall be maintained during traverse operations.

b. Primary means of helicopter movement around the yaw axis shall be through the application of differential tension in two tailguide cables attached to the airframe. Attachment and tension limits shall be in accordance with the criteria defined in SD-567-1.

c. Except for the physical attachment of the tailguide cables to the aircraft and the raising of the helicopter Tail Probe, all normal control of the RAST system during the straightening procedure and the traversing evolution shall be from the Control Console. The system shall provide for the sensing, transmission and display on the Control Console of the following physical conditions within the selected Rapid Securing Device.

Locked Arresting Beams at extreme starboard position
Locked Arresting Beams at extreme port position
Locked Arresting Beams at center position

d. A capabilitity to reel-out the tailguide cables shall be provided in which the force required at the on-deck ends to unspool the cables shall not exceed 75 lbs.

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e. The RAST system, through the Tailguide Winch, shall be capable of applying a cable tension of 3700 ± 300 lbs to either the port or starboard tailguide cable to swing the helicopter.

f. The RAST system shall provide tailguide cable tension of 85 ± 20 lbs under all operating conditions except where higher tensions are being commanded or when unspooling cable.

g. The RAST system, through the controlled application of power to the selected Traverse Winch, shall be capable of traversing the 20,000 lb SH-60B LAMPS MK III helicopter at speeds not exceeding 3.5 in/sec in "Low Traverse Speed" and not exceeding 13.2 in/sec in "High Traverse Speed". The design of the Traverse system shall include shock absorbing buffers at each end of traverse travel to limit the deceleration of the 20,000 lb aircraft traveling at 13.2 in/sec to not more than 0.2 g.

3.4.1.4 Launch. System operation in the Launch mode (see 3.1.1.4) shall provide the following capability within the stated limits.

a. Using controls located on the Control Console, the system shall provide for the unlocking and opening of the Rapid Securing Device Arresting Beams from around the aircraft Main Probe within two seconds of control actuation. Manual back-up control shall be provided by employing the lanyard described in 3.2.2.11.

3.4.2 Physical characteristics.

3.4.2.1 Health and safety. Health and safety criteria and all other aspects relating to the operating personnel shall be in accordance with MIL-STD-454 and MIL-STD-1472. A minimum safety factor of 2.5 shall be employed in the selection and use of wire rope.

3.4.2.2 Corrosion Protection

3.4.2.2.1 Material Selection - In selecting materials for the RAST system, consideration shall be given to the susceptability of the materials, individually and in combination with other materials, to the corrosive effects of the marine environment. Material combinations shall be selected to minimize the occurance of the galvanic corrosion.

3.4.2.2.2 Protective Measures.

3.4.2.2.2.1 General. Upon delivery, coating and paint work shall be complete and in a serviceable condition. Coated surfaces shall show no film failures (i.e., loss of adhesion, blistering, rusting, pinholding, checking, cracking, runs or sags) greater than 1 percent. Where film thickness is specified it shall be Dry Film Thickness (DFT), except as may otherwise be noted in this section.

For the purpose of classification of corrosion protective measures defined in the following paragraphs, equipment is classed as "sheltered or "exposed" as follows:

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Exposed - Rapid Securing Device
- Control Console
- Electric Cable Reel
- All traverse track components
- Above deck Tailguide Winch system components

Sheltered - All equipment not classed as exposed.

3.4.2.2.2.2 Application. A minimum of two coats shall be applied to satisfy all requirements for minimum film thickness except where one is specifically permitted herein. After the first coat has been applied, subsequent coats shall not be applied until the preceding coat has become properly dry and hard, unless otherwise specified herein. The dry film shall be free of surface contaminants including moisture. Before the application of any coat, all bare spots on the preceding coat shall be "touched-up".

Where the epoxy polyamide coating materials conforming to MIL-P-24441 are employed, each coat shall provide a DFT of 2 mils minimum and 4 mils maximum with the total three-coat system providing a DFT 8 mils minimum, 10 mils maximum.

If more than 1 week elapses between application of coats in an epoxy coating system, a thin, continuous "tack-coat" (1 to 2 mils wet) of the primer, or last coat applied, shall be applied prior to the application of next coat of the system.

3.4.2.2.2.3 Surface preparation (metal and plastic).

Surfaces to be painted, or treated with preservative, shall be completely free of corrosion products, mill scale, dirt, oil, grease, moisture, deteriorated paint, and other surface contaminiants. Coatings shall be applied as soon as practicable after cleaning, but in no instance longer than 24 hours.

a. Steel. Dirt and grease shall be removed as required. Rust and mill scale shall be removed from ungalvanized steel by blasting or other mechanical means. Where abrasive blasting is specified, steel surfaces shall be prepared to "near-White" metal in accordance with SURFACE PREPARATION SPECIFICATION NO. 10. (SSPC-SP-10) of the "Steel Structures Painting Council Manual, Systems and Specifications, Volume II, Section 2". Where specifications or designs requires bare steel, surfaces shall be coated with rust-protective compound of MIL-C-16173, Grade 2, or conforming to NATO Code C-620, to provide temporary protection for exposed surfaces. This coating shall be renewed as necessary to provide continued protection; the coating shall, however, be removed prior to placing assembled machinery into operation.

b. Aluminum. Aluminum surfaces shall be cleaned and prepared by light abrasive blasting or by power wire brushing, or orbital sanding. Only stainless steel bushes or scouring pads, or aluminum oxide abrasive sanding pads shall be used on aluminum surfaces. Steel shot (other than stainless steel), metallic slag, or metallic grit shall not be used for blast cleaning of aluminum surfaces. Blast cleaning materials for aluminum shall be aluminum oxide grit, mineral grit (iron free) or silica sand. Reclaimed abrasives which have been used to clean copper containing surfaces or other metals, shall not be used on aluminum surfaces.

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c. Where an inorganic zinc-coating has been used, surfaces shall be washed with high pressure fresh water, scrubbed, and rinsed with fresh water, to remove residual surface deposits, and then dried just prior to painting, or by the coating manufacturer's recommendations for removing such residual deposits.

d. Galvanized steel and miscellaneous metals. Galvanized steel, aluminum alloys, brass, copper, corrosion-resisting steel, and nickel copper alloys shall be cleaned to remove dirt, grease, corrosion products, and other surface contaminants prior to application of coating. Where blasting or mechanical tools are used, care shall be taken to only slightly roughen the surface to provide a good paint bond.

e. Painted metal surfaces. When painted surfaces show evidence of corrosion, peeling, blistering, checking, scaling, or general disintegration, paint shall be removed down to bare metal and surfaces recoated.

In touchup painting, where only localized areas or spots require repainting, removal of old paint shall be carried back around the edges of the spot or area until an area of complete intact and adhering paint film is present and no rust or blisters are evident. The edges of the remaining tightly adherent paint around the areas to be repainted shall be smoothed (feathered) down to the bare surface.

Where a surface is to be repainted and the old paint is not removed, the surface shall be clean and dry before new paint is applied.

f. Plastic surfaces. Surfaces requiring painting shall be lightly roughened. All extraneous matter shall be removed by wiping with a solvent and hand sanding, or by other means.

g. Hydraulic components. Non-ferrous and ferrous metal and alloyed pipe, tubing and fittings for hydraulic use shall be cleaned and protected in accordance with the provisions of MIL-STD-419.

3.4.2.2.2.4 Surface maintenance. After surface preparation, all machined, working or otherwide exposed surfaces of equipment and machinery, or components thereof, except those of nonferrous material, those of corrosion - resisting steel, or those to be immediately painted, shall be coated with a thin film of corrosion preventive conforming to MIL-C-16173. Grade of MIL-C-16173 coating employed shall be based on the material to be coated and the storage or handling anticipated during assembly or system build-up process until final preparation for delivery.

3.4.2.2.2.5 Painting. Surface of material shall receive pre-treatment, primer and finish coats as specified in the following sections. Where used, reference to Formula Nos. shall be interpreted in accordance with the following listing:

Formula No.	Name	Specification
84	Zinc chromate primer,	TT-P-645
111	Light gray alkyd	DOD-E-15090

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Formula No.	Name	Specification
116	Red lead primer, alkyd	DOD-P-17545
117	Wash primer pretreatment	DOD-P-15328
122-27	Gray, No. 27, vinyl alkyd	MIL-E-15836
150	Epoxy polyamide primer	MIL-P-24441
151	Epoxy polyamide, topcoat haze Gray	MIL-P-24441
156	Epoxy polyamide, Deck Red	MIL-P-24441

3.4.2.2.2.5.1 Machinery.

a. Unless specifically excepted, all external unmachined surfaces of ungalvanized ferrous metal (excpet CRES) parts of "sheltered" machinery shall be thoroughly cleaned and coated with (1) one coat of pre-treatment Formula 117 and one coat of primer Formula 84 or Formula 116, or (2) one coat of epoxy primer Formula 150. Two finish coats of enamel Formula 111, class 2, shall be applied except where thermal insulation is installed.

b. Internal and external surfaces of ungalvanized sheet metal work (except CRES) shall be given one coat of pre-treatment, Formula 117, one coat of primer, Formula 84 or 116, and finish coats to match the surrounding equipment. Heated internal surfaces of ungalvanized sheet metal work (except CRES), shall be given one coat of heat-resistant paint conforming to DOD-P-24555 over bare steel.

c. Unmachined external surfaces of ungalvanized ferrous castings may be finished smooth by filling with epoxy repair cement conforming to MIL-C-24176. Surfaces shall be cleaned by wire brushing, sanding, or other mechanical means to remove contaminants prior to application of the repair cement. Such parts shall then be given (1) one coat of pre-treatment Formula 117 and one coat of primer Formula 116, or (2) one coat of epoxy primer Formula 150, and (3) finished with one coat of Formula 111 class 2.

d. Painting of external surfaces of nonferrous metal casting, piping, or other parts is not required unless otherwise specified, except for parts of aluminum alloy. Aluminum alloy parts shall be given one coat of pre-treatment Formula 117 and one coat primer Formula 84 before assembly, after assembly, exposed surfaces shall be finish painted with 2 coats Formula 111, class 2.

3.4.2.2.2.5.2 Valves and piping.

a. The external surfaces of ungalvanized ferrous metal (except CRES) piping, fittings, and valves of "sheltered" equipment shall be cleaned and then painted with one coat of pre-treatment Formula 117, followed by one coat of primer Formula 84 or 116. Painting shall be accomplished before installation.

b. The external surfaces of ungalvanized (except CRES) piping, fittings, and valves of "exposed" equipment shall be abrasive blasted. After blasting, approximately 1-inch around circumference of pipe ends shall be masked-off. Three coats of epoxy ployamide system (one coat each, Formulas 150, 156, and 151), to a minimum dry film thickness of 8 mils shall be applied. If required, a finish coat to match the surrounding structure may be applied. Small fittings

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and valves may be hand or power cleaned to bare metal (blasting is not required) and coated with the epoxy coating system as specified above.

3.4.2.2.2.5.3 Miscellaneous. The following items shall not be primer coated or finish painted:

Cathodic protection anodes.
Glands, stems, yokes, toggle gear, and all machined external parts of valves.
Heat exchange surfaces.
Joint faces of gaskets and packing surfaces.
Lubricating gear, such as oil holes, oil or grease cups, lubricators, and surfaces in contact with lubricating oil.
Machined metal surfaces of reciprocating engines and pumps.
Name plates, label plates and information/warning plates.
Hydraulic and lubricating oil reservoirs, sumps or tanks.
Rods, gears, universal joints, and couplings of valve operating gear.
Rubber resilient elements of isolation mounts.
Sliding feet.
Springs.
Threaded parts.
Working surfaces.

3.4.2.2.2.5.4 Hydraulic equipment. Painting of the exterior surfaces of hudraulic equipment and components shall be in accordance with 3.4.2.2.2.5.1 through 3.4.2.2.2.5.3. Cleanliness and protective measures of MIL-STD-419 shall be maintained during and following the painting process.

3.4.2.2.2.5.5 Electric and electronic equipment. Electric and electronic equipment shall be painted as specified in this section. Unless otherwise specified, surface preparaton shall be in accordance with 3.4.2.2.2.3 followed by pre-treatment with one coat of Formula 117 and a primer coat of Formula 84.

a. Supplier coated items. Where topcoating is required for color or preservation (i.e. primer and finish coats), then coatings furnished by suppliers which do not conform to this requirement shall be finish painted as specified herein. If the finish coat specified herein is not compatible with the coating furnished by the equipment manufacturer, then the coating shall be removed, surfaces prepared, primed, and finish painted, as specified herein. Damaged areas shall be recoated as required for the original painting.

b. Finish coats. Except as othewise specified, electric and electronic equipment classed as "sheltered" shall be finished with two coats of Formula 111, class 2, and equipment classed as "exposed" shall be finished with two coats of silicone alkyd (Navy haze gray) conforming to TT-E-490, or two coats of Formula 122-27 if a vinyl system is used.

c. Unfinished surfaces. The following surfaces shall not be painted:

Gaskets and packing of spraytight and water-tight enclosures and fixtures.
Grounding contact surfaces.
Identification, operating, safety and warning label plates.

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Joints and contact surfaces of explosion proof enclosures.
Oil holes, grease cups and grease pipes of machinery.
External fasteners and assembly screws which are of contrasting color to the equipment on which they are used.

d. Metal enclosures. Enclosures for motor controllers, electric panels, wiring boxes, fittings, and fixtures (including those complying with Government standard drawings) and enclosures for electric and electronic equipment in general for which painting is not otherwise specified in Government specifications shall be coated in accordance with this section.

e. Motors and generators. Finish of motors and generators shall comply with MIL-E-917.

3.4.2.2.2.5.6 Equipment exposed to the weather. Surfaces of equipment exposed to the weather shall be cleaned and prepared in accordance with 3.4.2.2.2.3 prior to further coating in accordance with this section.

a. Steel. Unless coated in accordance with 3.4.2.2.2.6, steel surfaces which may be exposed to the weather (including the Rapid Securing Device, track trough components and tailguide sheave well components) shall receive one full coat of Formula 150, approximately 3 mils. This epoxy coating must be thoroughly cleaned and then coated with a thin "tack-coat" of Formula 150 prior to application of one coat of Formula 156 and a final coat of Formula 151. In applying Formulas 150 and 151 a film thickness in excess of 4-mils/coat shall be avoided.

Track plates and tail guide grid bars shall be given one full coat of Formula 150.

b. Aluminum. Aluminum surfaces shall be lightly blast cleaned and given a pre-treatment of chromate chemical conversion coating conforming to MIL-C-5541. The surfaces shall then be finish coated as specified above for steel.

3.4.2.2.2.6 Plating and special coating systems.

a. Cadmium plating. Cadmium plated parts, other than those approved under a formal parts control program of MIL-STD-965, shall not be employed.

b. Galvanized/inorganic zinc. The following items or materials shall not be galvanized or coated with inorganic zinc or heat applied galvanizing repair compounds:

(1) Structure, tanks and piping forming part of hydraulic system, lubricating oil or demineralized cooling water systems.

(2) Areas where electrical connection for installed equipment has been identified.

(3) Corrosion - resisting, high yield, steel

(4) Nonmagnetic steels other than Hadfield steel

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(5) Nonferrous metals other than copper-zinc alloys.

c. Ceramic coatings. Components to be installed in the RAST track trough, in the tailguide sheave wells and in the Rapid Securing Device, and which meet the following criteria, shall be coated with a ceramic, SermeTel (TM) or equal, coating. The criteria for selection of an assembly, sub-assembly or component shall include:

(1) The item is routinely exposed to a salt environment.

(2) The item is inaccessible for routine maintenance for extended periods of time.

(3) The item is compatible with the coating and curing process.

(4) The item, to the extent treated, is not subjected to abrasion or repeated working of the treated surface.

All track securing bolts and exterior surface of inserts shall be treated with the ceramic (SermeTel or equal) coating.

3.4.2.2.2.7 Anti-sieze, caulking and lubricating compounds. After assembly, all joints of dissimiliar metals exposed to the weather or seawater shall be sealed with a caulking compound conforming to MIL-C-18255 to prevent the entrance of moisture or water. Threaded parts shall be coated, before assembly, with anti-sieze mixture conforming to MIL-T-22361 or MIL-S-81733 depending on the anticipated frequency of disassembly.

Graphite lubricants shall not be utilized in the RAST system. Consideration shall be given to the use of anti-sieze compounds of MIL-T-22361 or a mixture consisting of equal parts by weight of 200 mesh zinc dust and MIL-S-8660 silicone compound in lieu of standard greases on equipments installed or used in the weather, to which access is limited, and which are particularly subject to galvanic corrosion.

3.4.3 Reliability.

3.4.3.1 Reliability program. The reliability program shall be conducted in accordance with MIL-STD-785. A reliability demonstration plan shall be developed to verify conformance with 3.4.3.2 and 3.4.3.3 (see 4.8.4).

3.4.3.2 Mission reliability. The mission reliability (see 6.5.8) shall equal or exceed a Mean Cycle Between Critical Failure (MCBFc) of 300 cycles. A cycle consists of the sequential operations of traversing the helicopter from the hangar to the launch position, a launch and subsequent recovery followed by straightening and traversing to the hangar position. For purposes of analyses and tests, equipment operating time for the nominal cycle is defined as 45 minutes consisting of 5 minutes for traverse from hangar to launch position, 15 minutes each for launch and recovery and 10 minutes for straightening and traverse to hangar.

3.4.3.3 MCBF (Mean Cycles Between Failure). MCBF, defined as the total number of cycles divided by the total number of failures, shall equal or exceed 50 cycles.

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3.4.3.4 Failure investigation, analysis, and corrective action system. A closed loop failure data collection, analysis, and corrective action system, in accordance with MIL-STD-785, para. 5.4.1, shall be developed and employed for all failures (see 4.7.6).

3.4.3.5 Failure reports. All relevant and nonrelevant failures shall be reported. A failure report containing the details of failure investigation shall be prepared (see 4.4.4), for each failure occurring in all contractually required testing.

3.4.4 Maintainability.

3.4.4.1 Maintainability program. The maintainability program shall be conducted in accordance with MIL-STD-470.

3.4.4.2 Maintenance downtime. The maintenance downtime values given in Table I shall not be exceeded when downtime is defined in accordance with MIL-STD-721 and occurs following a noncensored cycle (see 6.5.6, 6.5.7).

TABLE I. Maintenance Downtime

ITEM	Maintenance Support	Elapsed Maintenance Downtime (in hrs)	
	Direct Maintenance Manhours per cycle (see 6.5.2)	80th Percentile of Distribution	(MIL-STD-721) MTTR
Corrective maintenance of the system at organization level		3.70	1.00
Unscheduled (Corrective) Maintenance	.04		
Scheduled (Preventive) Maintenance	.36		

3.4.4.3 Maintenance Support. The maintenance support values of direct maintenance manhours per cycle at the organizational level shall not exceed 0.04 manhours per cycle for corrective maintenance and shall not exceed 0.36 manhours per cycle for preventive maintenance. These values shall be based on a system utilization rate of 1286 cycles per year and the assignment of not more than two maintenance personnel per maintenance event.

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3.4.4.4 Maintenance procedure. A complete maintenance procedure shall be developed using the support equipment and tools approved for organizational maintenance of the RAST System. The maintenance procedure shall be included in the technical manual to be prepared and delivered in accordance with the CDRL (see 6.3).

3.4.5 Environmental characteristics. The system shall be such that, when installed in accordance with the Installation Control Drawing, NAEC 700,000, all required functional characteristics are retained (including maintaining satisfactory lubrication and avoiding loss of oil from machinery or hydraulic systems) under the following shipboard conditions:

a. Temperature range of -28°C to +65°C for exposed equipment and 0°C to +50°C for sheltered equipment.

b. Exposure to a relative humidity of 95 percent for both continuous and intermittent periods, including conditions where condensation takes place on the equipment in the form of both water and frost.

c. When the ship is permanently trimmed down by the bow or stern as much as 5 degrees from the normal horizontal plane, is permanently listed as much as 15 degrees to either side of the vertical, is pitching 10 degrees up or down from its normal horizontal plane, or is rolling up to 45 degrees to either side of the vertical. Full system performance is not required when the ship roll exceeds 30 degrees; however, exposure to ship rolling conditions up to 45 degrees to either side of the vertical shall not cause loss of capability when rolling is reduced to 30 degrees or less.

3.4.5.1 Shock. The Rapid Securing Device shall conform to Grade A shock requirements of MIL-S-901 for those conditions without an aircraft load on the securing device. All other equipments shall conform to Grade B shock requirements of MIL-S-901.

3.4.5.2 Vibration. Unless otherwise specified, systems and system components shall be qualified in accordance with Type I of MIL-STD-167-1.

3.4.5.3 Salt fog. The Rapid Securing Device, Control Console, Electric Cable Reel and interconnection box for the Electric Cable Reel shall withstand a salt fog test specified in 4.8.3.6.

3.4.5.4 Fungus. Equipment items employing electric or electronic technology shall be resistant to fungi in both the operating and non-operating condition and shall not be adversely affected when tested in accordance with 4.8.3.7.

3.4.5.5 Dust. The Rapid Securing Device, Control Console and Electric Cable Reel shall withstand the dust (fine sand) test of 4.8.3.10.

3.4.5.6 Enclosures. Housings for electrical and electronic circuits, parts and components shall conform to the degree of enclosure requirements of MIL-STD-108. Electrical assemblies in the Rapid Securing Device and the enclosure of the Control Console shall be watertight; all other electrical enclosures shall be dripproof (45 degrees).

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3.5 Design and construction.

3.5.1 Parts control. Selection and use of parts during design and fabrication of RAST systems shall be from those parts contained in a Program Parts Selection List (PPSL) developed and approved in accordance with MIL-STD-965, Procedure I. Peculiar parts, as defined in MIL-STD-965, shall be exempt from parts control.

3.5.1.1 Standard Electronic Modules. Where employed, the selection, design and use of standard electronic modules shall be in accordance with MIL-STD-1378.

3.5.1.2 Interchangeability. All replaceable items in the system design carrying the same part number shall be interchangeable (see 6.5.9).

3.5.2 Electromagnetic radiation and interference. Reduction of electromagnetic radiation shall be in accordance with MIL-STD-461. Techniques to reduce generated electromagnetic interference shall be in accordance with MIL-STD-1310.

3.5.3 Identification. System nomenclature and individual item identification plates shall be in accordance with MIL-N-18307.

3.5.4 Equipment installation provisions. RAST system equipment shall conform to the installation requirements of Installation Control Drawing, NAEC 700,000. The equipment shall be supplied with suitable drilled mounting brackets, resilient pads, special adjustable braces, and any bearing or sliding blocks.

3.5.5 Electrical.

3.5.5.1 Termination. All units shall facilitate inter-unit wiring. Provisions shall be included to comply with the General Specifications for Ships of the U.S. Navy, sections 304, 305, and 400.

3.5.5.2 Cables and connectors. Construction and electrical interfaces shall be in accordance with MIL-STD-1683; electric cabling shall conform to MIL-C-915; connectors shall be selected and employed as per MIL-STD-1353. Equipment shall also be supplied with both mating halves of each connector.

3.5.6 Hydraulic. All units shall facilitate inter-unit connection. Intra-unit piping shall terminate at a between-unit bulkhead fitting, appropriately identified for interface connection. Design for hydraulic piping, both inter-unit and intra-unit, shall employ o-ring face seal connectors and other provisions conforming to MIL-STD-777. Inter-unit piping shall not be supplied unless specifically stated as part of the contract. Equipment shall be supplied with both mating halves of each coupling.

3.5.6.1 Sampling and bleed valves. Sampling valves shall be James, Pond and Clark Co. Part Number P14-500, or equal, conforming to MIL-V-81940. Bleed valves shall be Fluir Bleed Valve Co. Part Number 7579K, or equal, conforming to MIL-V-81940.

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3.5.6.2 Filters. Filters shall employ disposable element filters fitted with differential pressure indicators conforming to MIL-F-24402.

3.5.7 Safety.

3.5.7.1 Operational. The RAST system design shall be guided by the concept that component or subsystem failure will not place the aircraft, ship, personnel or the remainder of the system in jeopardy.

3.5.7.2 Danger marking. Any area hazardous to personnel during operation of the equipment shall be distinctively identified in conformance with MIL-STD-1472. Suitable warning signs shall be located in close proximity to describe the danger.

3.5.7.3 Foreign object damage (FOD). Equipment to be located on the flight deck shall be designed to minimize the susceptibility for collection of debris that could be ingested by aircraft engines or rotor blades causing damage and/or personnel hazard. Entrapment areas shall be easily inspected and cleaned.

3.6 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.2.3).

3.7 Test reports. Test reports shall be prepared in accordance with MIL-STD-831 covering all classifications of inspection specified in 4.2. Delivery of test reports shall be as specified in the CDRL (see 6.3).

3.7.1 Refurbishment data. Complete records shall be kept on equipment assemblies and components thereof that are to be delivered as spares or as part of a deliverable system that are refurbished to meet acceptance criteria (see 4.8.1).

3.8 Workmanship. Workmanship shall conform to requirement 9 of MIL-STD-454. Hydraulic assemblies piping, hoses and connectors shall be free of leaks under all required operating conditions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality control system. The contractor shall provide and maintain an effective quality control system acceptable to the Government covering the supplies and services being procured. Unless otherwise specified (see 6.2), the quality program shall be in accordance with MIL-Q-9858 and when specified, a quality assurance plan shall be prepared as specified (see 6.3).

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4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practice, methods, and procedures of the written system are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government inspection of delivered items to assure compliance with this specification and as specified in the contract or order (not excluding any requirement of the specification for which detailed tests are not specified herein).

4.1.3 Failures. During developmental, qualification or acceptance testing conducted by, or under the supervision of the contractor, records shall be maintained of each failure, test conditions, operating time and other pertinent details and reported in the applicable test report (see 3.7 and 4.4). In addition, the modifications and improvements implemented to correct the deficiencies shall be fully documented and reported (see 4.4.4).

4.2 Classification of inspection. The methods of inspection fall within the following classifications:

a. Initial performance tests
b. Operational burn-in
c. First article inspection
d. Quality conformance inspection
   (1) Production inspection
   (2) Production control inspection

4.2.1 Initial performance test. After completion of manufacturing, assembly and production adjustment operations, and prior to proceeding with any of the other inspections of 4.2, each item shall be operated under the standard ambient condition (see 4.6) and a record made of all data necessary to determine compliance with required system performance (see 3.4.1). These data shall provide the criteria for checking satisfactory performance of the item either during or at the conclusion of operational burn-in.

4.2.2 Operational burn-in. After completion of initial performance tests each item shall be operated for a period of time or a number of cyclic operations, under stressing conditions sufficient to expose infant mortality type failures. The duration employed shall be based on the susceptibility to infant mortality of the technology employed in the item. The contractor shall include the proposed burn-in duration and stressing conditions in the test plan prepared in accordance with 4.3. The duration of the burn-in for each item shall be subject to further adjustment during production, with procurement agency approval, to obtain optimum effectiveness commensurate with cost.

4.2.3 First article inspection. The item(s) designated for first article inspection as specified (see 6.2), shall be subjected to all examination and tests necessary to determine compliance with this specification. Unless otherwise specified (see 6.2), first article inspection shall include examination and test shown in table II and shall be performed as approved in accordance with 4.3.

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4.2.4 Quality conformance inspection.

4.2.4.1 Production inspection. A production inspection shall be made on each item offered for delivery. This inspection shall include such examination and tests as will prove the workmanship and reveal omissions and errors of the production process such as, functional and performance test at a limited number of points in the required range, tests which detect deviations from design, tests of controls/adjustments, and tests which detect hidden defects of materials. Unless otherwise specified (see 6.2), production inspection shall include the examination and tests shown in table II and shall be performed as approved in accordance with 4.3.

4.2.4.2 Production control inspection. Production control inspection shall be conducted on a sampling basis when specified in the contract or order (see 6.2). This inspection shall consist of the examination and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. When ordered (see 6.2), production control inspection shall include the examination and tests shown in table II and shall be performed as approved in accordance with 4.3.

4.3 Examination and test plans. The contractor shall submit written examination and test plans which contain the procedures, data forms, a list of test equipment to be used and the sequence of test for performing operational burn-in, first article and quality conformance inspections to the procuring activity via the Government inspector for approval prior to beginning inspection as specified in the CDRL (see 6.3). Examination and tests shall be identified in accordance with 4.2.

4.3.1 Test procedures. Test procedures shall contain step-by-step information for each inspection, test and demonstration to be performed. The test procedures shall include preparation for test, connections, adjustments, measurements, including test parameters and tolerance limits, and disconnection of item under test.

4.3.1.1 Test equipment listing. The test equipment list shall identify the test equipment to be used by name, model, range and accuracy. Special test equipment shall also be listed and identified as to its capability to correctly perform the tests.

4.3.1.2 Test data forms. Test data forms shall be prepared for each of the procedures. These forms shall be organized to indicate the acceptance test limits of the test and the actual readings observed. In addition, the form shall indicate a statement of acceptance or rejection after each test. When test data forms are specified by the referenced specification, they shall be utilized. The form shall contain a signature and date block for certification of the data by the test engineer.

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4.4. Test reports.

4.4.1 Initial performance and operational burn-in reports. Test data sheets for initial performance tests and for operational burn-in tests shall be included in the test reports of 4.4.2 or 4.4.3 as applicable for the specific units tested.

4.4.2 First article inspection report. A first article inspection report containing results of all tests shall be prepared as specified in the CDRL (see 6.3). When the tests are performed in accordance with referenced documents, the test data and test data forms shall comply with these referenced documents.

4.4.3 Quality conformance inspection reports. Quality conformance inspection reports containing the results of all tests performed shall be prepared as specified in the CDRL (see 6.3). The report shall list the serial numbers of the equipment tested, and the date tested. When tests are performed in accordance with referenced documents, the test data and test data forms shall comply with these referenced documents.

4.4.4 Failure reporting. Any failure occurring during or as a result of tests specified herein shall be included in the reports of 4.4.2 and 4.4.3, together with the probable cause of failure and corrective action taken.

4.5 Test facilities, chambers, and apparatus. Test facilities, chambers, and apparatus used in conducting the tests specified herein shall be capable of meeting all of the applicable test conditions. The accuracy of instruments and test equipment used to control the test parameters, whether located at a Government testing laboratory, at the contractor's plant or at an independent testing facility shall satisfy the requirements of MIL-C-45662 for calibration cycle of specific instruments. Records of calibration history shall be maintained and the latest calibration date shall be attached to each test instruments. All instruments and test equipment used in conducting the tests specified herein shall:

a. Conform to laboratory standards whose calibration is traceable to a calibration source in accordance with MIL-C-45662;
b. Have an accuracy no worse than one-third the tolerance for the parameters to be measured. In the event of conflict between this accuracy and a requirement for accuracy in any one of the specific test methods, the specific test method shall govern;
c. Be appropriate for measuring the test parameters.

4.6 Standard test conditions. Unless otherwise specified herein or in the individual referenced standards, all measurements and tests shall be made under the following standard ambient conditions:

a. Temperature 23° ± 10° C (73° ± 18° F).

b. Relative humidity 50 ± 30 percent

c. Atmospheric pressure 725^{+50}_{-75} mm ($28.5\ ^{+2.0}_{-3.0}$) inches of mercury.

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TABLE II. Examination and tests.

Subject	Requirement	Test Method	First Article	Quality Conformance Inspection	
				Production Inspection	Production Control Inspection
Examination					
Weight	3.3.2.1	4.8.1(a)	X	X	
Size	3.3.2.1	4.8.1(b)	X	X	
Parts & Material	3.5	4.8.1(c)	X	X	
Finish	3.4.2.2	4.8.1(d)	X	X	
Marking	3.5.3 & 3.5.7.2	4.8.1(e)	X	X	
FOD	3.5.7.3	4.8.1(f)	X	X	
Workmanship	3.8	4.8.1(g)	X	X	
Intial Performance		4.2.1	X	X	
Burn-in		4.2.2	X	X	
Performance					
Full System	3.4.1	4.8.2.1	X		X
Individual Item	3.2.2 & 3.4.1	4.8.2.2	X	X	
Environmental					
Electrical	3.3.2.3	4.8.3.1	X		X
Temperature	3.4.5	4.8.3.2	X		X
Vibration	3.4.5.2	4.8.3.3	X		X
Shock	3.4.5.1	4.8.3.4	X		
Humidity	3.4.5	4.8.3.5	X		
Salt Fog	3.4.5.3	4.8.3.6	X		
Fungus	3.4.5.4	4.8.3.7	X		
Inclination	3.4.5	4.8.3.8	X		
Enclosure	3.4.5.6	4.8.3.9	X		X
Dust	3.4.5.5.	4.8.3.10	X		
Reliability Demonstration	3.4.3	4.8.4	X		X
Maintainability Demonstration	3.4.4	4.8.5	X		

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4.7 Performance of test.

4.7.1 Installation of test item. The test item shall be installed in the test facility in a manner that will simulate service usage making connections and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation, but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. For tests where temperature values are controlled, the test chamber shall be at standard ambient conditions when the test item is installed. The test item shall then be operated to determine that no malfunction or damage was caused due to faulty installation or handling. The requirement for operation following installation of the test item in the test facility is applicable only when operation is required during or immediately following exposure to the specific test.

4.7.2 Performance measurement. All performance measurements shall be made with instruments of the accuracy specified in 4.5.

4.7.3 Pre-performance test. Prior to proceeding with any of the test methods, the test item shall be operated under the standard ambient conditions (see 4.6) and a record made of all data necessary to determine compliance with required performance. These data shall provide the criteria for checking satisfactory performance of the test item either during or at the conclusion of the test or both, as required.

4.7.4 Performance check during test. When operation of the test item is required during the test exposure, the operation and performance checks shall be of sufficient duration or shall be repeated at appropriate intervals to insure a record of comprehensive comparative data for comparison with data recorded under standard ambient conditions.

4.7.5 Visual examination. Prior to, during, and after each test, the test item shall be visually examined and a record made of any damage or deterioration. If a test chamber is used for the test, a visual examination of the test item within the chamber shall be performed at test conditions when possible.

4.7.6 Failure criteria. Unless otherwise specified herein, equipment, or portions thereof, subjected to a test specified herein shall be considered to have failed the test when any of the following occur:

a. Performance parameters exceed limits specified.
b. Catastrophic or structural failure.
c. Distortion or displacement of mechanical parts that causes difficulty of servicing or replacing a part.
d. Any condition that results in a hazard to personnel safety.
e. Deterioration, corrosion, or change in performance limits causing failure to meet operational service or maintenance requirements.
f. Deterioration of sealing compounds that would cause a decrease in service life or reliability.
g. Any condition that requires or results in a corrective maintenance action.

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4.7.7 Failure investigation, analysis, and corrective action. A closed loop failure data collection, analysis, and corrective action system, in accordance with MIL-STD-785, paragraph 5.4.1, must be employed for all failures.

4.8 Test methods.

4.8.1 Surface examination. The equipment shall be examined to ensure compliance with the applicable referenced specifications with respect to:

a. Weight (see 3.3.2.1).
b. Size (see 3.3.2.1)
c. Parts and materials (see 3.5).
d. Finish (see 3.4.2.2)
e. Marking (see 3.5.3 and 3.5.7.2)
f. FOD (see 3.5.7.3)
g. Workmanship (see 3.8)

In the process of examination, the item shall not be disassembled in a manner that will affect the performance, durability or appearance of the item.

4.8.2 Performance test. The equipment shall be subjected to a performance test including all operating controls to determine compliance with the performance parameters and safety requirements herein.

4.8.2.1 Full system performance test. System equipment, installed in accordance with established installation and interface requirements and combined into a system configuration, shall be subjected to a performance test to determine compliance with 3.4.1; tests shall be performed as approved in accordance with 4.3.

4.8.2.2 Individual item performance test. Individual equipment items shall be subjected to a functional and performance test at a limited number of points in the required operating range to detect deviations from design. Operating test points shall include the full range of operating controls/adjustments installed in the item. The test shall be performed as approved in accordance with 4.3.

4.8.3 Environmental tests. All equipments shall successfully pass environmental testing as specified in the following paragraphs. Equipment for which documentation exists to demonstrate its capability to meet the equivalent or more severe environmental requirement may be exempt from a specific test with concurrence of the procuring activity. Item environmental test requirements are specified in Table III.

4.8.3.1 Electrical tests. Control and power system elements shall be subjected to following electrical tests in accordance with MIL-E-16400.

a. Steady state voltage and frequency
b. Transient voltage
c. Transient frequency (control system only)
d. Power interruption

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TABLE III. Environmental test requirements.

Test Method	Test Items
4.8.3.1 Electrical Tests	All control & power system elements
4.8.3.2 Temperature Tests	All operating equipments
4.8.3.3 Vibration Test	All operating equipments
4.8.3.4 Shock Test	All operating equipments
4.8.3.5 Humidity	Rapid Securing Device, Control Console (CC), Electric Cable Reel (ECR), Motor Starters (MS), Relay Interconnecting Box (RIB), Electric Component Assembly (ECA), Interconnecting Boxes (IB)
4.8.3.6 Salt Fog	RSD, CC, ECR, IB for ECR
4.8.3.7 Fungus	CC, ECR, MS, RIB, ECA
4.8.3.8 Inclination	RSD
4.8.3.9 Enclosure tests	CC, RIB, ECA, RSD, MS, ECR, IB
4.8.3.10 Dust	CC, ECR, RSD

4.8.3.2 Temperature tests. Equipment shall undergo temperature tests in accordance with MIL-STD-810, method 501.1, procedure I, and method 502.1, procedure I, except the high/low storage temperature portion of each test shall be deleted, to determine compliance with 3.4.5. The Rapid Securing Device and Control Console are defined as exposed equipment; all others are defined as sheltered.

4.8.3.3 Vibration test. Equipment shall undergo the Type I vibration testing of MIL-STD-167-1 (see 3.4.5.2).

4.8.3.4 Shock test. Equipment shall undergo shock testing per MIL-S-901 to demonstrate compliance with 3.4.5.1. Testing of the Rapid Securing Device shall be conducted on a section of the track, but with no aircraft loads imposed.

4.8.3.5 Humidity. Humidity tests shall be conducted on applicable equipment items in accordance with Method 507.1, Procedure IV of MIL-STD-810 except that operation and measurements at the end of the fifth cycle shall be conducted at a stablized temperature of + 50°C (see 3.4.5).

4.8.3.6 Salt fog. Salt fog test shall be conducted on applicable equipment items in accordance with Method 509.1, Procedure I of MIL-STD-810. Electrical operation is required at the end of the 48 hour exposure period (see 3.4.5.3).

4.8.3.7 Fungus. Fungus test shall be conducted on applicable equipment items in accordance with Method 508.1 of MIL-STD-810. Electrical operation is required at the end of the test period (see 3.4.5.4).

4.8.3.8 Inclination. Inclination tests shall be conducted on the RSD to verify performance of all functions at inclinations up to ± 30°. In addition,

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the RSD shall be exposed to inclinations up to ±45° to verify that such exposure will not cause loss of capability when inclination is reduced to ±30° or less. These angles are required only for rotation of the RSD's horizontal plane about its longitudinal axis (see 3.4.5).

4.8.3.9 Enclosure tests. Degree of enclosure tests of MIL-STD-108 shall be conducted to determine conformance of applicable equipment items to requirements. Testing of the RSD and Control Console for watertight integrity shall be by exposure to nozzle-directed water flow rather than immersion (see 3.4.5.6).

4.8.3.10 Dust. Dust (fine sand) test shall be conducted on applicable equipment items in accordance with Method 510.1, Procedure I of MIL-STD-810. Operation of test items during test exposure is not required. Specified performance shall be obtained during operation at Step 5 (see 3.4.5.5).

4.8.4 Reliability demonstration. The contractor shall conduct a reliability demonstration in accordance with the criteria of Table IV on one of the first three RAST production systems. Demonstration testing shall be conducted subsequent to satisfactory completion of full system performance tests, shall be conducted on equipment installed in a system configuration, and shall be conducted in accordance with the reliability demonstration plan approved in accordance with 4.3.

TABLE IV. Reliability demonstration criteria

Parameter	Required	Allowable Failures	Test Cycles
MCBFc	300	3	1,750
MCBF	50	25	1,750

4.8.4.1 Mission profile. The demonstration shall begin with the RAST System in the condition of full operating capability after satisfactorily completion of a pre-operational inspection and an operational readiness test. The test mission profile shall consist of the following elements or their test equivalents.

a. Traverse from hangar to launch area.
b. Launch
c. Recovery to securing
d. Straightening
e. Traverse from recovery area to hangar.

4.8.4.2 Mission success criteria. The mission success criteria constitutes the functions required of each system for successful mission completion. The failure of any equipment resulting in the inability of the system to perform a mission required function during any phase of a test cycle in which that function is required shall constitute a failed mission cycle; such failure shall be considered a critical failure in the computation of MCBFc (see 3.4.3.2).

4.8.4.3. Cycle success criteria. Initiation of the next cycle with the system in the condition of full operating capability without corrective maintenance between cycles constitutes a successful cycle. Any relevant failure (see 4.7.6 and 4.8.4.6) shall be counted in the computation of MCBF (see 3.4.3.3).

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4.8.4.4 Performance check. A full system performance test, as approved in the demonstration test plan, shall be performed on the system before and after the reliability demonstration testing, as a minimum.

4.8.4.5 Maintenance. Preventive maintenance shall be limited to that identified in the demonstration test plan. Suspicion of incipient failure shall not be justification for replacing a part prior to its actual failure. However, if the failure of this part may cause catastrophic failure, it may be replaced with the approval of the procuring activity. A part so replaced shall be treated as a failed item and for the purposes of maintenance data collection and maintainability calculations, the catastrohpic failure shall be assumed to have occurred. All preventive maintenance and corrective maintenance actions shall be recorded as demonstration data.

4.8.4.6 Failure classification. All failures shall be classified as relevant or nonrelevant in accordance with MIL-STD-2074.

4.8.5 Maintainability demonstration. The contractor shall conduct a formal maintainability demonstration to assure the maintainability requirements of this specification. The demonstration shall be performed in accordance with MIL-STD-471A, Test Method 8 on one of the first three RAST production systems, utilizing maintenance procedures, support equipment and tools approved for use with the system. Corrective maintenance that occurs as a result of failures during other First Article Testing may be used toward this demonstration requirement if performed in accordance with approved maintenance procedures and properly documented.

4.9 Disposition of tested equipment

4.9.1 Refurbishment. Equipment assemblies and components thereof that are to be delivered as spares or as part of a deliverable system shall be refurbished to meet production acceptance criteria. All restoration actions shall be fully documented (see 3.7.1).

4.9.2 Non-repairable equipment. Disposition of equipment deemed nonrepairable by the contractor shall be as approved by the procuring activity.

5. PACKAGING

5.1 RAST system. Unless otherwise specified, (see 6.2) preservation, packaging, packing and marking for shipment of all procured RAST Systems shall be performed in accordance with MIL-STD-794, Level "C".

5.2 Spares.

5.2.1 Preparation. All designated spare assemblies or subassemblies shall be functionally tested, "burned-in" and adjusted where required to allow direct item replacement. Items that require in-system fine tuning shall be conspicuously marked with an appropriate instructional placard.

5.2.2 Packaging and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing and marking for shipment of all procured spares shall be performed in accordance with MIL-STD-794 Level "A".

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6. NOTES

6.1 Intended use. The RAST system to be supplied in accordance with this specification is intended to provide the capability to operate LAMPS MK III helicopters in adverse weather conditions from the small flight decks of LAMPS MK III ships experiencing extreme motions.

6.1.1 General LAMPS mission statement. A requirement exists to provide the capability to operate helicopters in all weather conditions, up to and including upper Sea State 5, from the decks of nonaviation ships operating within the LAMPS mission profile.

6.1.2 RAST mission profile. It is anticipated that helicopter recovery operations aboard LAMPS ships will be conducted in the following manner. The helicopter will approach from astern of the ship, flying to arrive in a hover position at a safe altitude above the flight deck. After the messenger cable has been lowered to within reach, the ship's personnel ground the cable and attach the messenger link to the recovery assist cable which had previously been faked out on the deck. The messenger cable will be reeled up causing the recovery assist cable to lock into the main probe of the helicopter. The recovery assist cable is then reeled in until taut whereupon the control system automatically maintains a constant hover tension. To optimize the guiding effect of the cable, the hover altitude is decreased to an acceptable minimum and the cable tension is increased as desired for the existing environmental conditions. During a quiescent period, the LSO assists the pilot to land and actuates the RSD to capture the helicopter main probe. The pilot actuates the tail probe which engages the tail grid. Following the usual helicopter post flight operations, the tailguide winch cables are manually attached, the tail probe is retracted, and the recovery assist cable is released. Using a combination of tailguiding and traversing controls, the helicopter is maneuvered to the required position for traversing. With the tailguide winch cables removed, the helicopter can be traversed into the hangar.

6.1.2.1 Launch operations. After the helicopter has been traversed to the takeoff position and readied for flight, the tail probe may be retracted leaving the helicopter secured to the RSD by the main probe alone. When suitable conditions for takeoff are obtained, the LSO operates the console control to disengage the RSD and the helicopter is free to takeoff.

6.1.2.2 Untethered operations. When ship motions are not too extreme and with due consideration to other environmental and atmospheric factors, the pilot may land without the assistance of the recovery assist cable. During untethered operations, the LSO guides the pilot so that the helicopter lands with the main probe inside the RSD; subsequent actions are then the same as for tethered landings.

6.1.3 Free deck operations. Although outfitted with a recovery system, LAMPS ships will also conduct flight operations with unequipped helicopters. In this situation, the RSD will be traversed away from the landing area.

6.2 Procurement requirements. Procurement documents shall specify the following:

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a. Title, number and date of this specification.
b. First article inspection requirements (see 3.6, 4.2.3 and 6.4).
c. Quality program requirements if other than in accordance with MIL-Q-9858 (see 4.1.1).
d. Any deviation from production inspection requirements (see 4.2.4.1).
e. Items, or sampling basis, to be selected for production control inspections if required (see 4.2.4.2).
f. Identification of specific composition of system elements required for systems procured for different classes of ships. See tabular listing of composition and quantities in NAEC Drawing 700,000.
g. Level of preservation, packaging and packing if other than Level "A" for spares or Level "C" for systems.

6.3 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations, the data requirements identified below will be developed in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.4.3.5, 4.1.3 and 4.4.3	Failure Report	UDI-R-21141A
3.4.4.4	Technical Manual	UDI-M-21368
3.5.1	Program Parts Selection List	DI-E-7027
3.5.3	Request, Nomenclature and Identification Assignment and Approval	UDI-E-21336A
4.1.1	Quality Assurance Program Plan	UDI-R-21374A
4.2.2 and 4.3	Test Plan (Burn-In)	DIT-5204
4.2.3 and 4.3	First Article Qualification Test Plan	DI-T-5315A
4.2.4.1 and 4.3	Acceptance Test Procedures (Production Inspection)	DI-T-3714A
4.4	First Article Test Report	UDI-T-23790
4.2.4.1 and 4.4.	Acceptance Test Report (Individual Unit Production Inspection)	DI-T-3721A
4.4	Test Report (Production Control Inspection)	DI-T-2072

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(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations. The first article should be a single representative system configuration composed of sample individual items selected from the first three production systems. Individual item tests under the first article examination may be conducted on sample items selected from the first three production systems. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.5 Definitions. For purposes of this specification, definitions in MIL-STD-109, MIL-STD-280, MIL-STD-721 and the following apply.

6.5.1 Operational readiness test. That set of functional tests and observations necessary for the operating crew to determine the operational usability of the RAST system.

6.5.2 Direct maintenance manhours. Direct maintenance manhours (DMMH) are the number of direct labor manhours expended on the RAST System or related support equipment, including operation of the support equipment. Manhours for the following indirect activities shall not be included in DMMH: alert duty, supervision, administration, management, training, meetings, work load control, material control, travel, and any other activity inherently associated with delay time.

6.5.3 Organizational maintenance. Organizational maintenance includes all maintenance performed at, on, in, or to the RAST System using only those skills, tools, support equipment, publications, procedures, and techniques specified and approved for normal service use when deployed.

6.5.4 Product. Includes parts, subassemblies, assemblies, units, equipments and sets.

6.5.5 Full operating capability. The state wherein all equipment is functional and operating within specified limits.

6.5.6 Non-censored cycle. An attempted demonstration cycle that is accountable for reliability and maintainability purposes as determined by the Test Board before the cycle occurs.

6.5.7 Censored cycle. An attempted demonstration cycle that is not accountable for reliability and maintainability purposes.

6.5.8 Mission reliability. Mission reliability is the probability that RAST System can perform all mission functions successfully.

6.5.9 Interchangeable. The characteristics of an item which (1) possesses such functional and physical characteristics as to be equivalent in performance, reliability, and maintainability, to another item of similar or identical

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purposes; and (2) is capable of being exchanged for the other item (a) without selection for fit or performance and (b) without alteration of the items themselves or of adjoining items, except for adjustment.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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